INCH-POUND

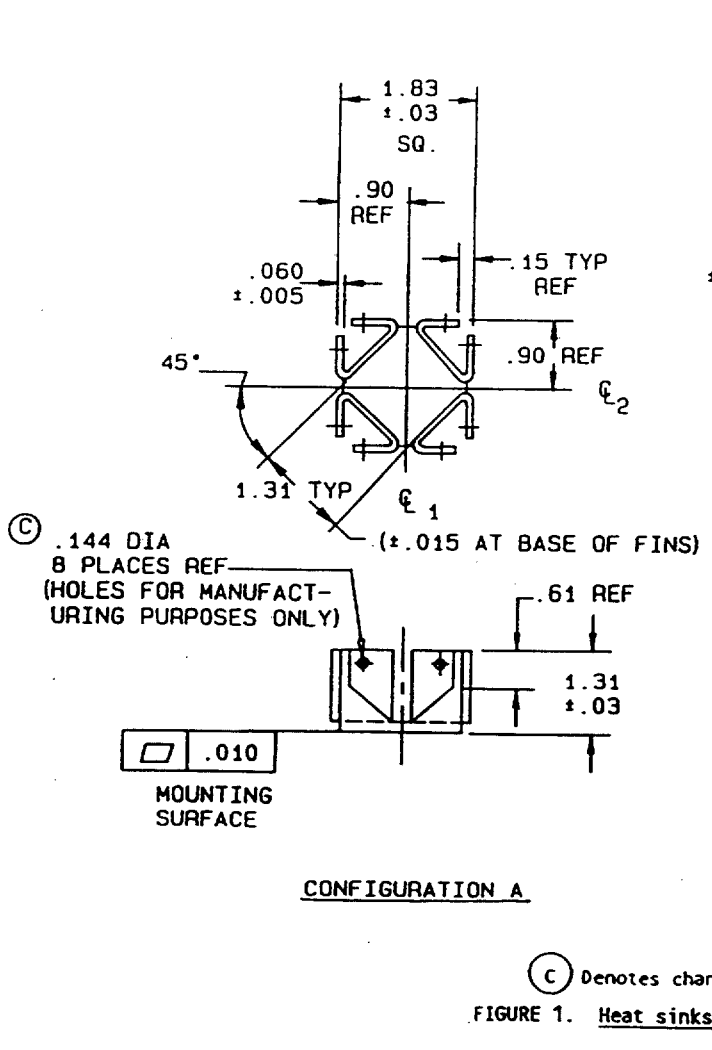
MIL-H-87111/5C
11 Dec 1992
SUPERSEDING
MIL-H-87111/5B
18 April 1986

MILITARY SPECIFICATION SHEET

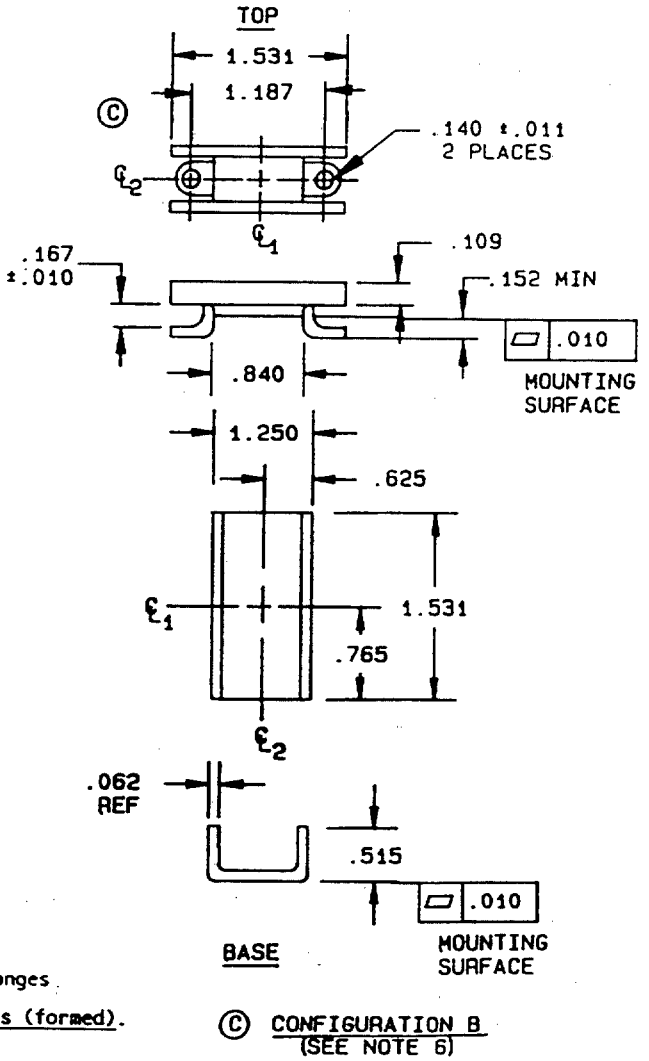
HEAT SINK, ELECTRICAL-ELECTRONIC COMPONENT, SEMICONDUCTOR DEVICE, CLASS 1 (FORMED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-H-87111.



CONFIGURATION A



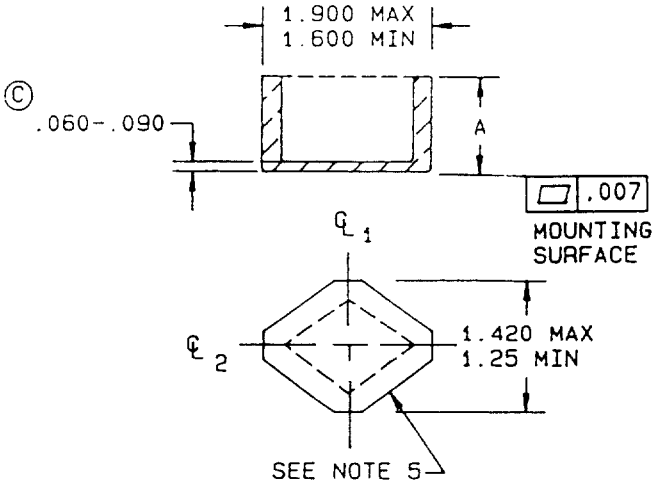
© CONFIGURATION B
(SEE NOTE 6)

© Denotes changes

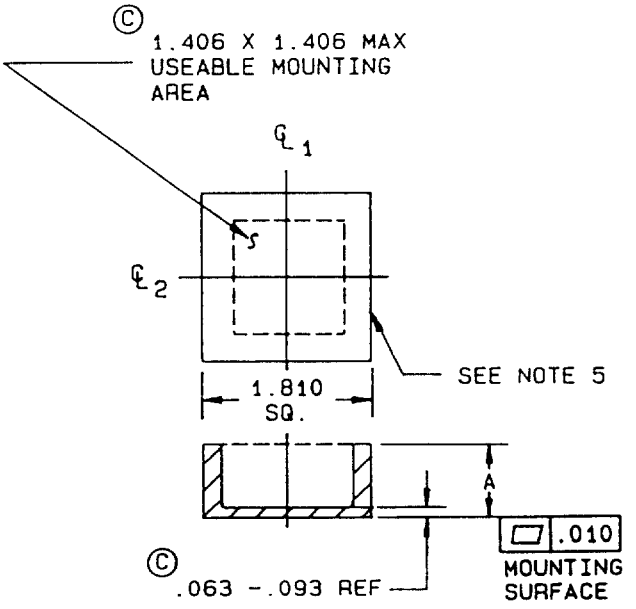
FIGURE 1. Heat sinks (formed).

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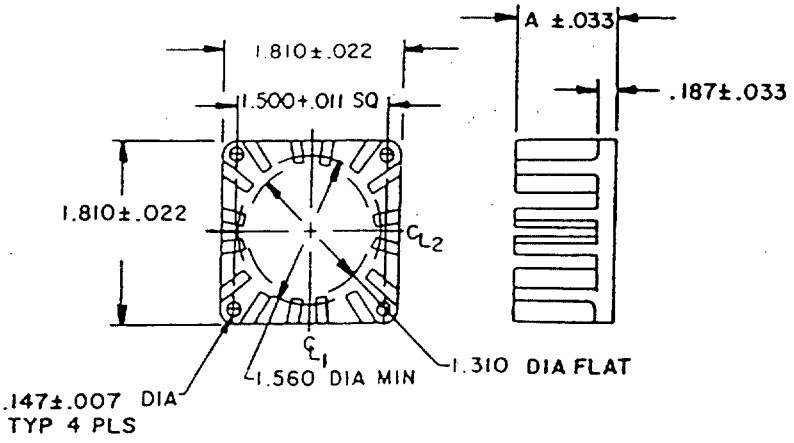
CONFIGURATION C



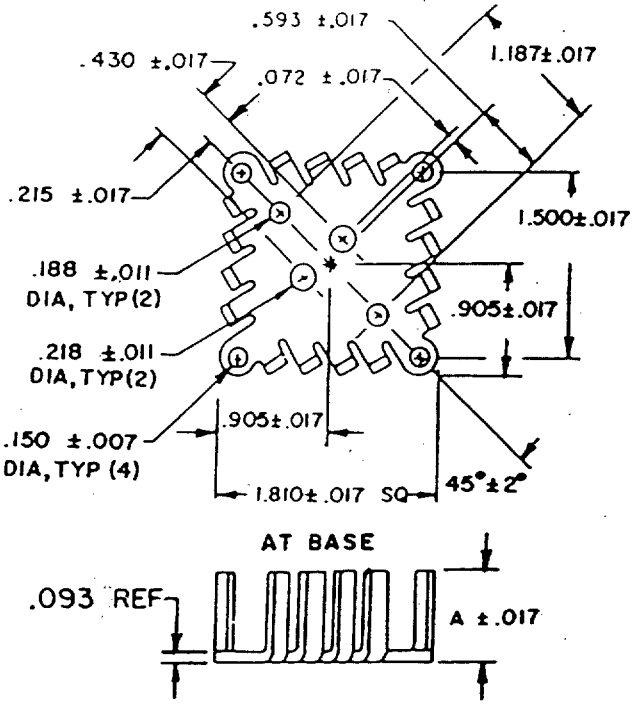
CONFIGURATION D

FIGURE 1. Heat sinks (formed) - Continued.

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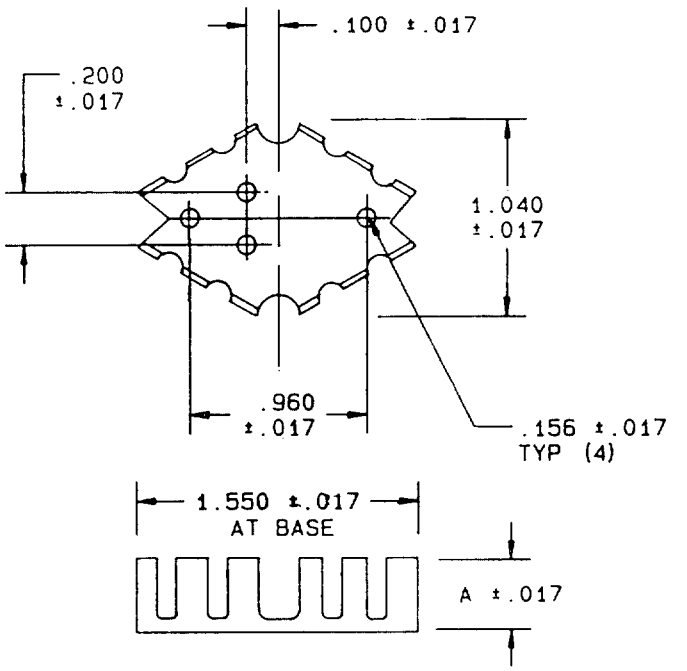
CONFIGURATION E



CONFIGURATION F

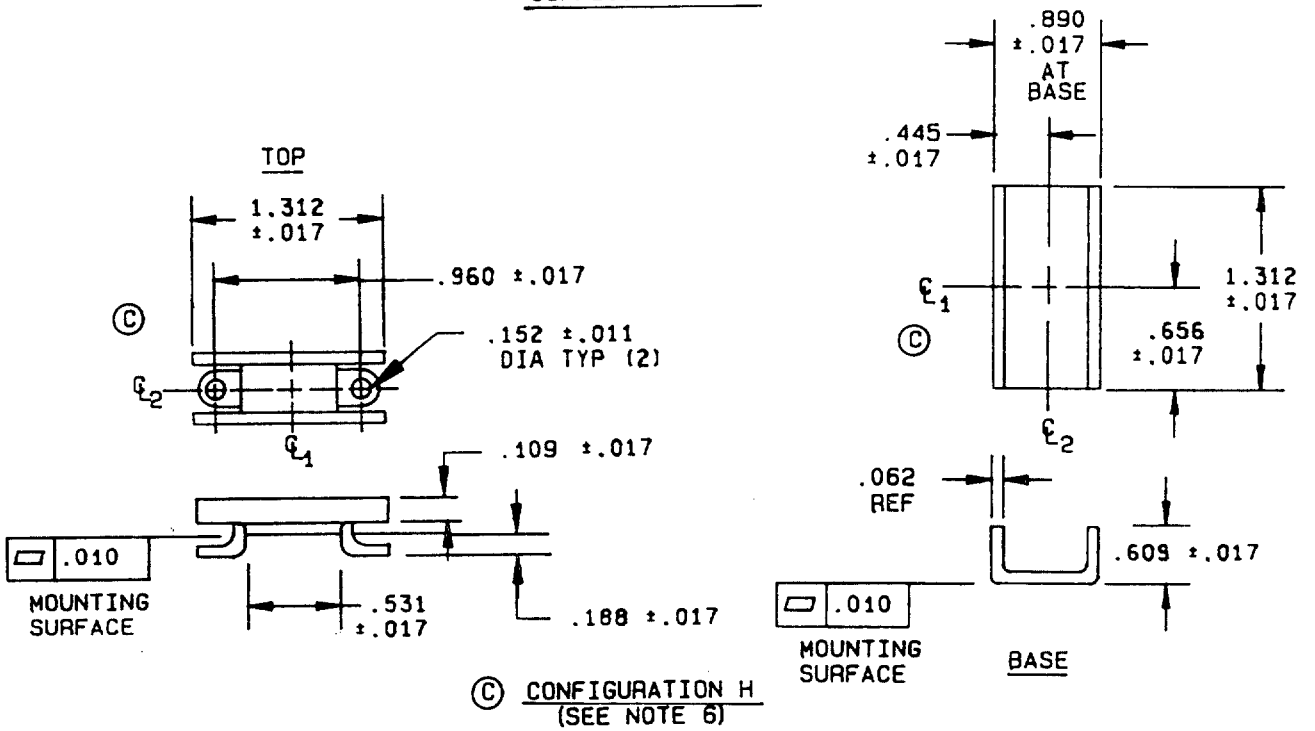
FIGURE 1. Heat sinks (formed) - Continued.

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Ⓒ MATERIAL THICKNESS .062

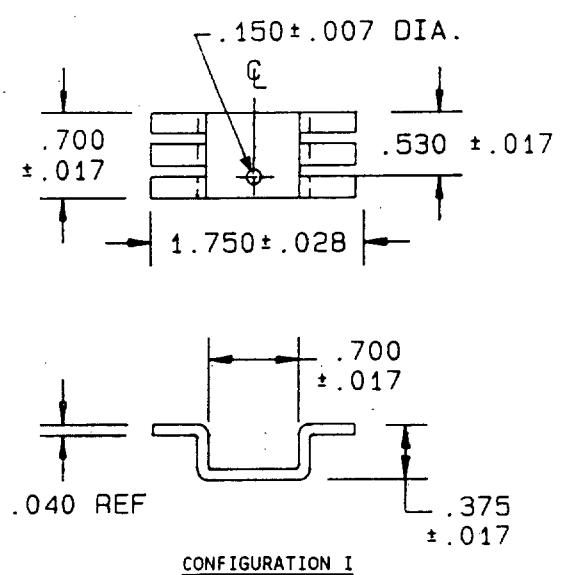
CONFIGURATION G



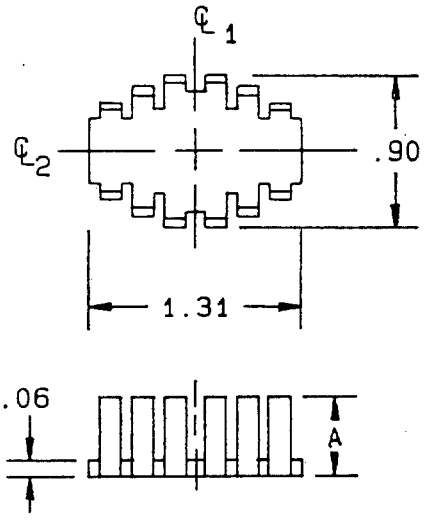
Ⓒ CONFIGURATION H
(SEE NOTE 6)

FIGURE 1. Heat sinks (formed) - Continued.

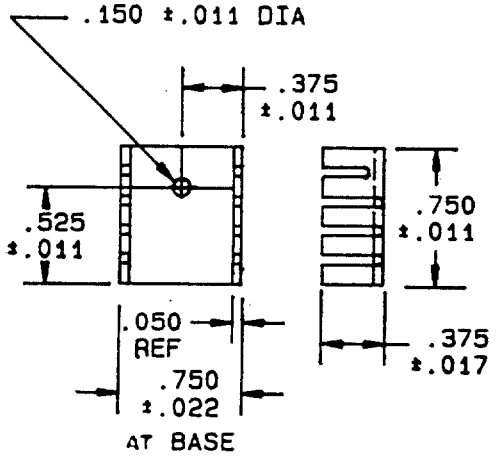
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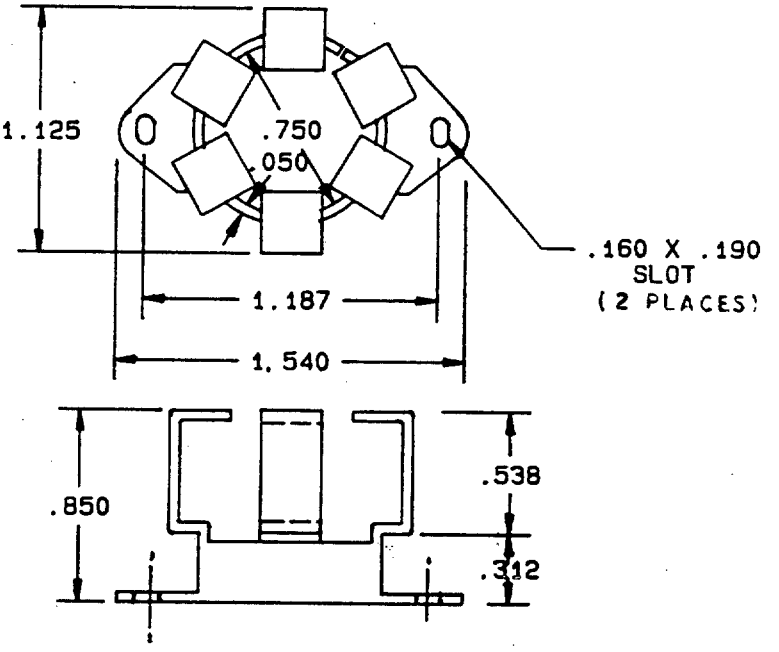
CONFIGURATION I



CONFIGURATION K



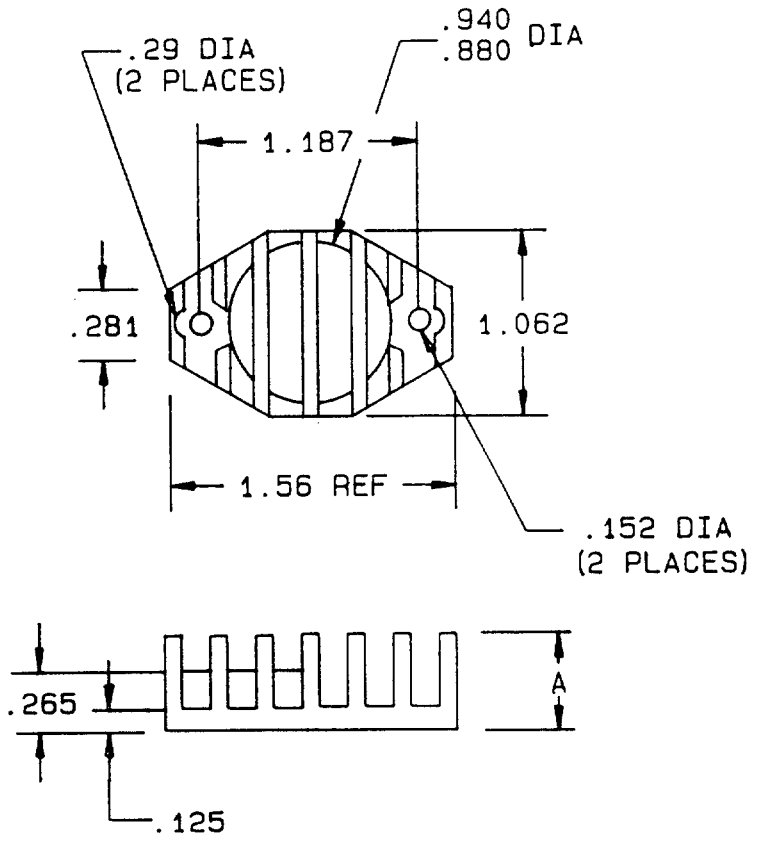
CONFIGURATION J



© CONFIGURATION L

FIGURE 1. Heat sinks (formed) - Continued.

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(C) CONFIGURATION M

FIGURE 1. Heat sinks (formed) - Continued.

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Inches	mm	Inches	mm	Inches	mm
.005	0.13	.187	4.35	.890	22.63
.007	0.18	.188	4.78	.90	22.9
.010	0.25	.190	4.83	.905	22.99
.011	0.28	.200	5.08	.940	23.88
.015	0.38	.218	5.54	.960	24.38
.017	0.43	.25	6.3	1.040	26.42
.022	0.56	.265	6.73	1.062	26.98
.03	0.8	.281	7.14	1.09	27.7
.033	0.84	.312	7.93	1.1	27
.04	1.0	.375	9.52	1.125	28.57
.040	1.02	.406	10.31	1.187	30.15
.050	1.27	.430	10.92	1.25	31.7
.06	1.5	.445	11.30	1.250	31.75
.060	1.52	.500	12.70	1.31	33.3
.062	1.57	.515	13.08	1.310	33.27
.063	1.59	.525	13.34	1.312	33.32
.072	1.83	.530	13.46	1.400	35.56
.09	2.3	.531	13.49	1.420	36.06
.093	2.36	.538	13.68	1.500	38.10
.100	2.54	.545	13.84	1.531	38.89
.109	2.77	.609	15.47	1.540	39.12
.125	3.18	.61	15.5	1.550	39.37
.140	3.56	.625	15.88	1.56	39.6
.144	3.66	.656	16.66	1.560	39.62
.147	3.73	.700	17.78	1.600	40.64
.15	3.8	.750	19.05	1.750	44.45
.150	3.81	.765	19.43	1.810	45.97
.152	3.86	.840	21.34	1.83	46.5
.166	3.96	.850	21.59	1.880	47.75
.160	4.24	.880	22.35	1.900	48.26
.167	4.24				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.017 (0.43 mm) on decimals and ±2° on angles. A .062 (1.57 mm) radius is permissible on corners and edges.
(C) 4. All configurations: Diagrams depict general designs only. Individual manufacturers' designs may vary slightly within the dimensions shown.
(C) 5. Configurations C and D only: Fin design (staggered, in-line, etc.) is optional within the dimensions and thermal resistance specified.
(C) 6. Configurations B and H only: Top and base are delivered as an assembly.

FIGURE 1. Heat sinks (formed) - Continued.

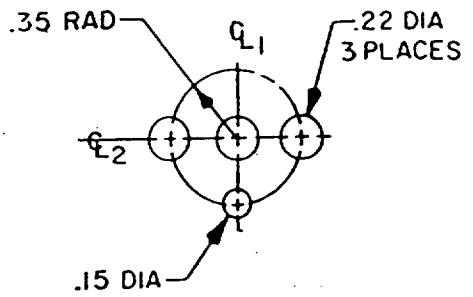
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© TABLE I. Part or Identifying Number (PIN).

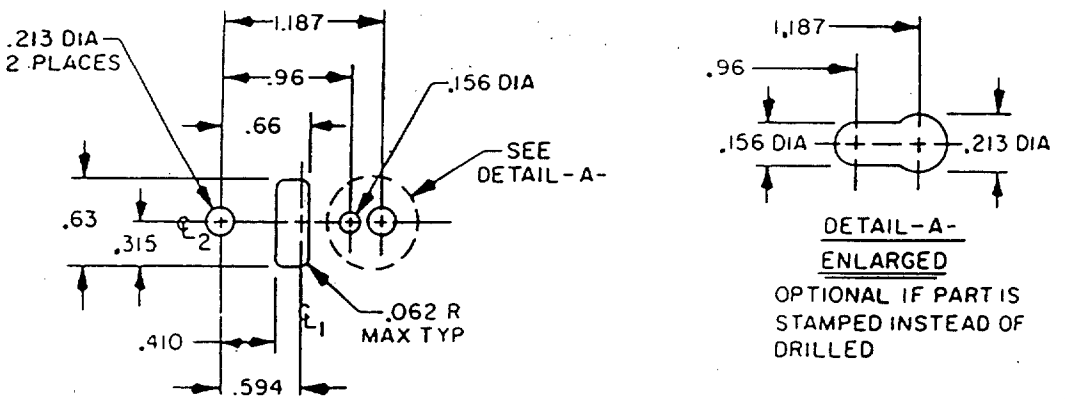
PIN M87111/05- 1/																					
Hole pattern (see figure 2) 2/																				As shown on figure 1	Configuration	Dimension A (±.020)	3/ 4/ Thermal resistance at 75°C rise
H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13	H14	H15	H16	H17	H18	H19					
1-001	1-011	1-021	1-031	1-041	1-051	1-061	1-071	1-081	1-091	1-101	1-111 5/	1-121 5/	1-131	1-141	1-151	1-161	1-171	1-181 5/					
1-002 5/	1-012	1-022 5/	1-032 5/	1-042 5/	1-052	1-062	1-072	1-082	1-092 5/	1-102 5/	1-112 5/	1-122 5/	1-132 5/	1-142 5/	1-152 5/	1-162 5/	1-172	1-182 5/					
1-003 5/	1-013	1-023	1-033	1-043	1-053	1-063	1-073	1-083	1-093	1-103 5/	1-113 5/	1-123 5/	1-133 5/	1-143	1-153	1-163	1-173	1-183 5/					
1-004 5/	1-014	1-024	1-034	1-044	1-054	1-064	1-074	1-084	1-094	1-104 5/	1-114 5/	1-124 5/	1-134 5/	1-144	1-154	1-164	1-174	1-184 5/					
1-005 5/	1-015	1-025	1-035	1-045	1-055	1-065	1-075	1-085	1-095	1-105 5/	1-115 5/	1-125 5/	1-135 5/	1-145	1-155	1-165	1-175	1-185 5/					
1-006 5/	1-016	1-026	1-036	1-046	1-056	1-066	1-076	1-086	1-096	1-106 5/	1-116 5/	1-126 5/	1-136 5/	1-146	1-156	1-166	1-176	1-186 5/					
1-007	1-017	1-027	1-037	1-047	1-057	1-067	1-077	1-087	1-097	1-107	1-117	1-127	1-137	1-147	1-157	1-167	1-177	1-187					
1-008	1-018	1-028	1-038	1-048	1-058	1-068	1-078	1-088	1-098	1-108	1-118	1-128	1-138	1-148	1-158	1-168	1-178	1-188					
1-009	1-019	1-029	1-039	1-049	1-059	1-069	1-079	1-089	1-099	1-109	1-119	1-129	1-139	1-149	1-159	1-169	1-179	1-189					
1-010	1-020	1-030	1-040	1-050	1-060	1-070	1-080	1-090	1-100	1-110	1-120	1-130	1-140	1-150	1-160	1-170	1-180	1-190					
1-191	1-200	1-209	1-218	1-227	1-236	1-245	1-254	1-263	1-272	1-281	1-290	1-299	1-308	1-317	1-326	1-335	1-344	1-353					
1-192	1-201	1-210	1-219	1-228	1-237	1-246	1-255	1-264	1-273	1-282	1-291	1-300	1-309	1-318	1-327	1-336	1-345	1-354					
1-193	1-202	1-211	1-220	1-229	1-238	1-247	1-256	1-265	1-274	1-283	1-292	1-301	1-310	1-319	1-328	1-337	1-346	1-355					
1-194	1-203	1-212	1-221	1-230	1-239	1-248	1-257	1-266	1-275	1-284	1-293	1-302	1-311	1-320	1-329	1-338	1-347	1-356					
																			1-362 2/	F	.500 (12.70)	7.2°C/W	
																			1-363 2/	F	.750 (19.05)	6.1°C/W	
																			1-364 2/	F	1.000 (25.40)	5.2°C/W	
																			1-365 2/	F	1.250 (31.75)	4.2°C/W	
																			1-366 2/	G	.500 (12.70)	12.2°C/W	
																			1-367 2/	G	.750 (19.05)	7.9°C/W	
																			1-368 2/	G	1.000 (25.40)	6.7°C/W	
1-195 5/	1-204 5/	1-213 5/	1-222 5/	1-231 5/	1-240 5/	1-249 5/	1-258 5/	1-267 5/	1-276	1-285 5/	1-294 5/	1-303 5/	1-312	1-321	1-330	1-339	1-348 5/	1-357 5/		H	---	14.5°C/W	
																			1-369 2/	I	---	16.66°C/W	
																			1-370 2/	J	---	21.4°C/W	
1-196 5/	1-205 5/	1-214	1-223	1-232 5/	1-241 5/	1-250 5/	1-259 5/	1-268 5/	1-277	1-286 5/	1-295 5/	1-304 5/	1-313 5/	1-322	1-331	1-340	1-349 5/	1-358 5/		K	.250 (6.35)	19.7°C/W	
1-197 5/	1-206 5/	1-215	1-224	1-233 5/	1-242 5/	1-251 5/	1-260 5/	1-269 5/	1-278	1-287 5/	1-296 5/	1-305 5/	1-314 5/	1-323	1-332	1-341	1-350 5/	1-359 5/		K	.500 (12.70)	19.7°C/W	
1-198 5/	1-207 5/	1-216	1-225	1-234 5/	1-243 5/	1-252 5/	1-261 5/	1-270 5/	1-279	1-288 5/	1-297 5/	1-306 5/	1-315 5/	1-324	1-333	1-342	1-351 5/	1-360 5/		K	.750 (19.05)	19.7°C/W	
1-199 5/	1-208 5/	1-217	1-226	1-235 5/	1-244 5/	1-253 5/	1-262 5/	1-271 5/	1-280	1-289 5/	1-298 5/	1-307 5/	1-316 5/	1-325	1-334	1-343	1-352 5/	1-361 5/		K	1.000 (25.40)	19.7°C/W	
																			1-371	L	---		
																			1-372	M	.500 (12.70)		
																			1-373	M	.750 (19.05)		
																			1-374	M	1.000 (25.40)		

© 1/ Complete PIN will include a letter to indicate finish (see PIN requirement).
2/ The hole pattern for dash numbers 1-362 through 1-370 shall be as shown on figure 1.
3/ Thermal resistance in natural convection.
4/ Thermal resistance values shown are guidelines and are intended only for comparisons between the various configurations.
© 5/ This dash number deleted with no replacement.
© 6/ Configurations D and E only: Mounting hole patterns H2, H6, H7, H8, H9, and H18 shall be rotated 45° off centerline.

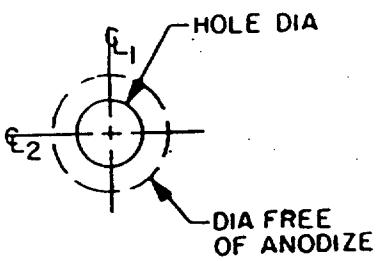
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H1 (TO-36 JEDEC CASE, TO-6 JEDEC CASE)



H2 (UNIVERSAL, TO-3, TO-66, TO-126 AND TO-220)

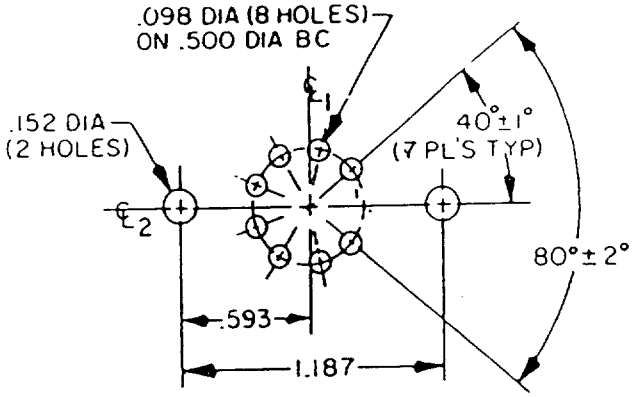


Hole Diameter	Diameter free of anodize ±.035	Hole pattern number
.200 ±.005	.500	H3
.270 ±.007	.750	H4
.330 ±.007	1.000	H5

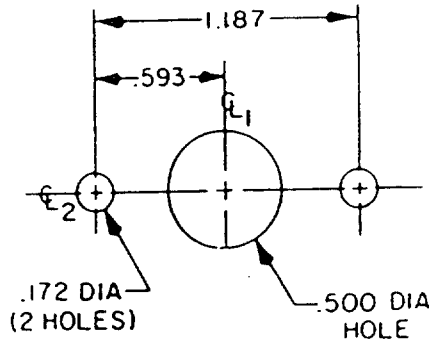
H3, H4, AND H5 (STUD)

FIGURE 2. Mounting hole patterns

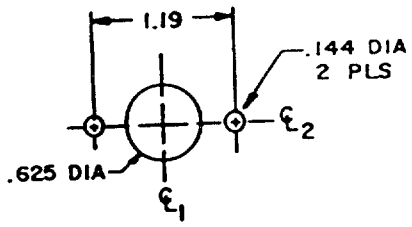
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H6 (TO-3 1C JEDEC case)



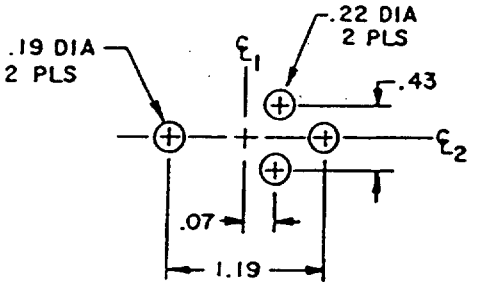
H7 (TO-8 1C JEDEC case)



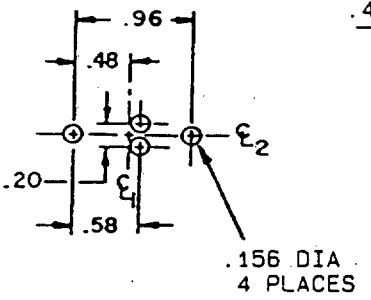
H8 (MULTI-LEAD TO-3)

FIGURE 2. Mounting hole patterns - Continued

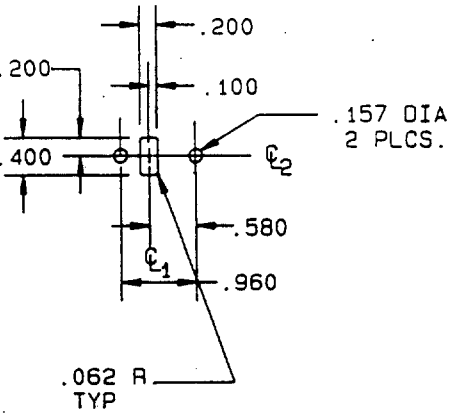
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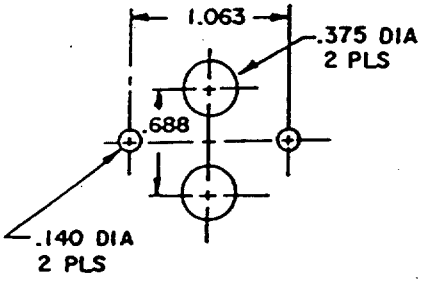
H9 (TO-3)



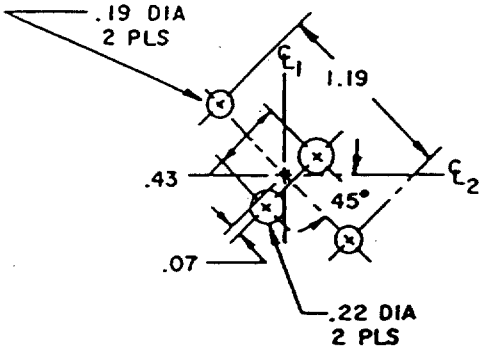
Ⓒ H10 (TO-66)



Ⓒ H10 (optional pattern for stamped parts)



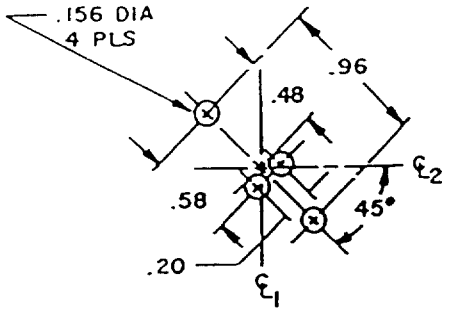
H11 (TWO TO-220's)



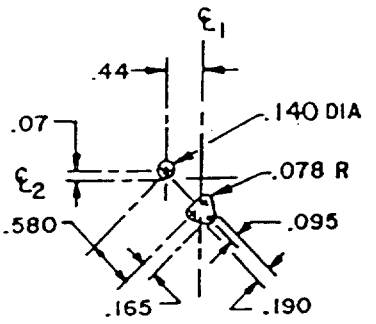
H12 (TO-3)

FIGURE 2. Mounting hole patterns - Continued.

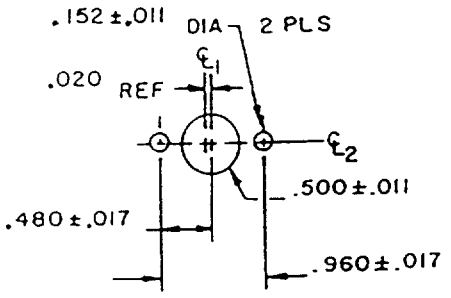
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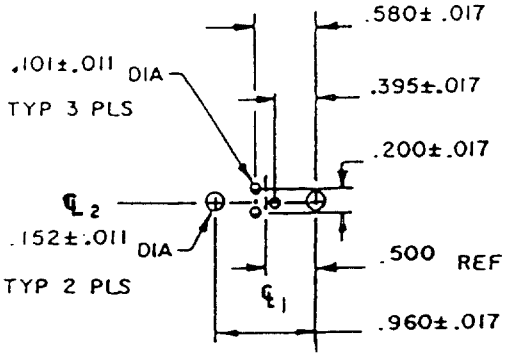
H13 (TO-66)



H-14 (TO-220)



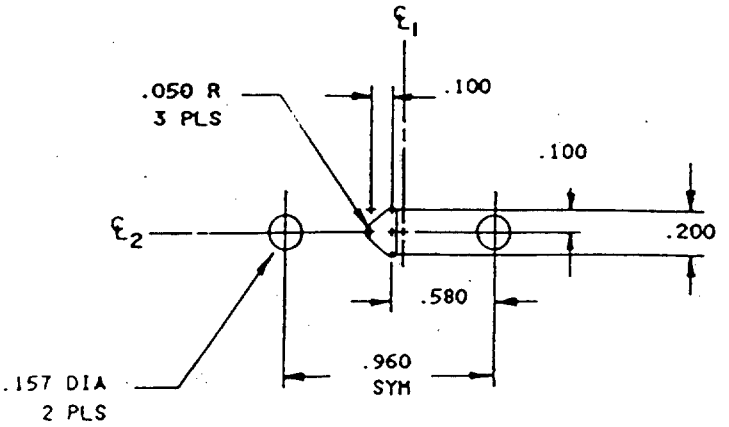
H15 (TO-66)



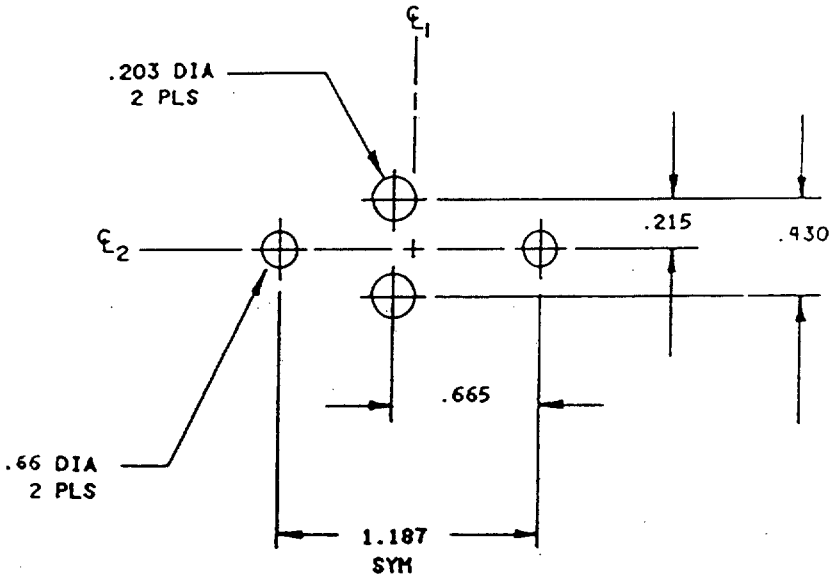
Ⓒ H-16 (TO-66)

FIGURE 2. Mounting hole patterns - Continued.

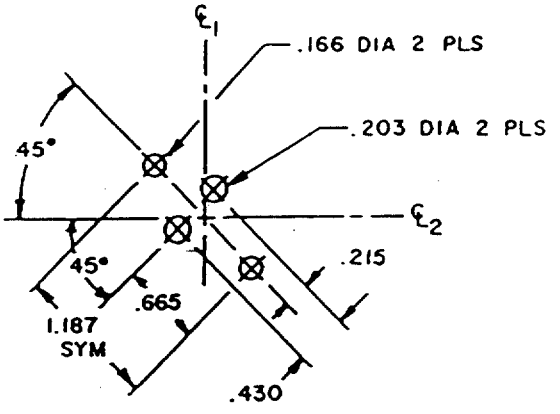
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H17 (TO-66)



H18 (TO-3)



H19 (TO-3)

FIGURE 2. Mounting hole patterns - Continued.

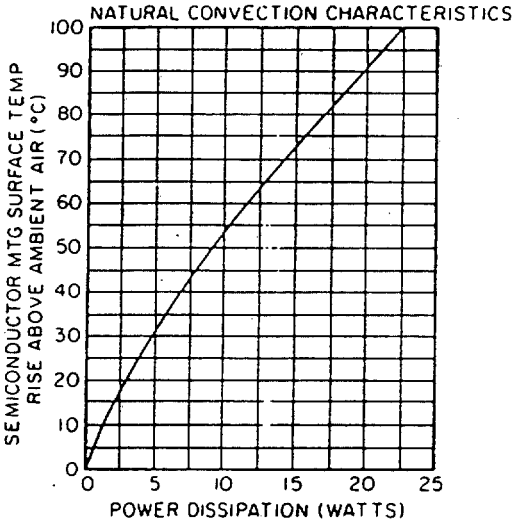
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Check the source to verify that this is the current version before use.

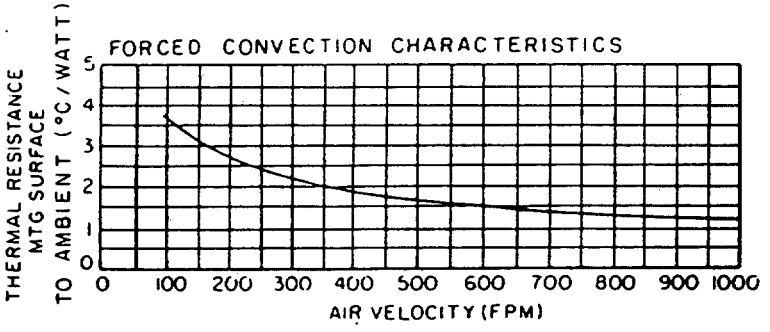
Inches	mm	Inches	mm	Inches	mm
0.005	0.13	.166	4.22	.48	12.2
0.007	0.18	.172	4.37	.480	12.19
0.011	0.28	.19	4.8	.500	12.7
0.017	0.43	.190	4.82	.58	14.7
0.020	0.51	.20	5.1	.580	14.73
0.035	0.89	.200	5.08	.593	15.06
0.050	1.27	.203	5.16	.594	15.09
0.062	1.57	.213	5.41	.625	15.88
0.07	1.78	.215	5.46	.63	16.0
0.078	1.98	.22	5.6	.66	16.8
0.095	2.41	.250	6.35	.665	16.89
0.098	2.49	.270	6.86	.688	17.48
0.100	2.54	.315	8.00	.750	19.05
0.101	2.57	.330	8.38	.96	24.4
0.140	3.56	.35	8.9	.960	24.38
0.144	3.66	.375	9.53	1.000	25.40
0.15	3.8	.410	10.41	1.063	27.00
0.152	3.86	.43	10.9	1.187	30.15
0.156	3.96	.430	10.92	1.19	30.2
0.157	3.99	.44	11.2	1.250	31.75
0.165	4.19				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for decimals and ±2° for angles.
4. A .062 (1.57 mm) radius is permissible on corners and edges.

FIGURE 2. Mounting hole patterns - Continued.

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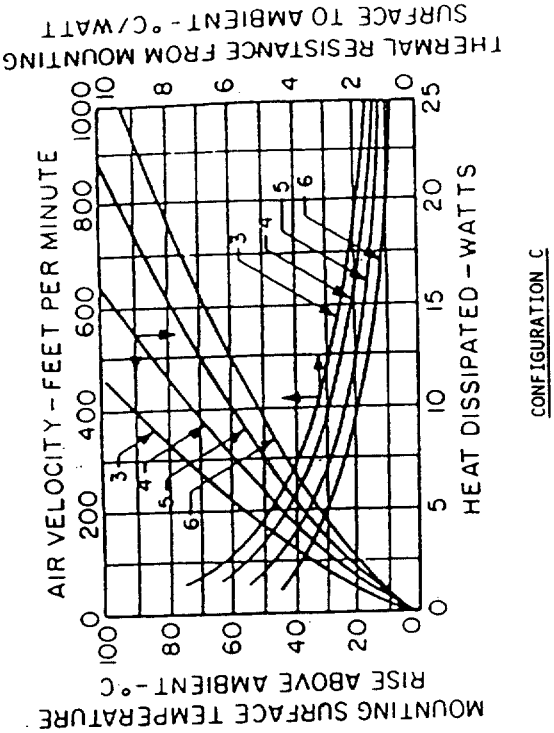




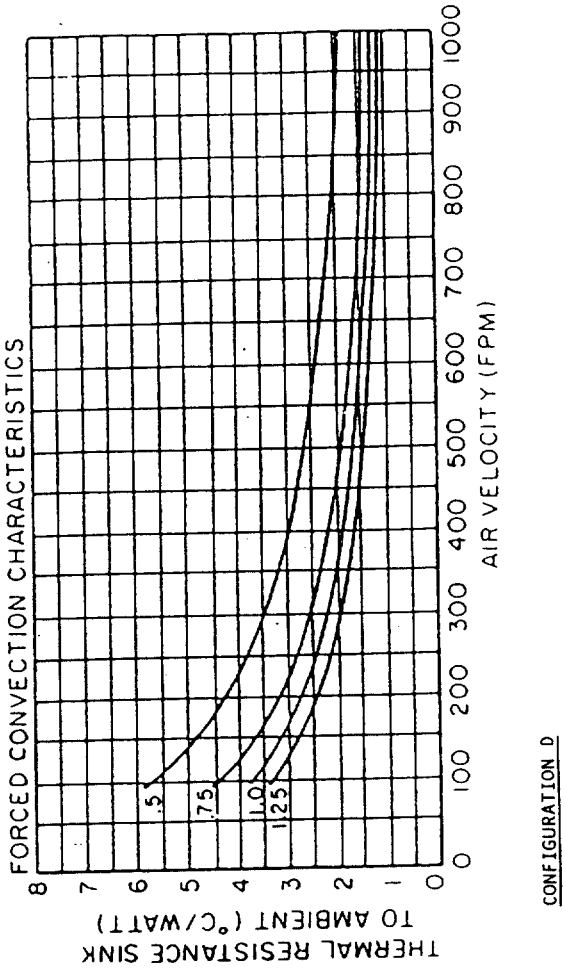
CONFIGURATION A

FIGURE 3. Thermal performance characteristics.

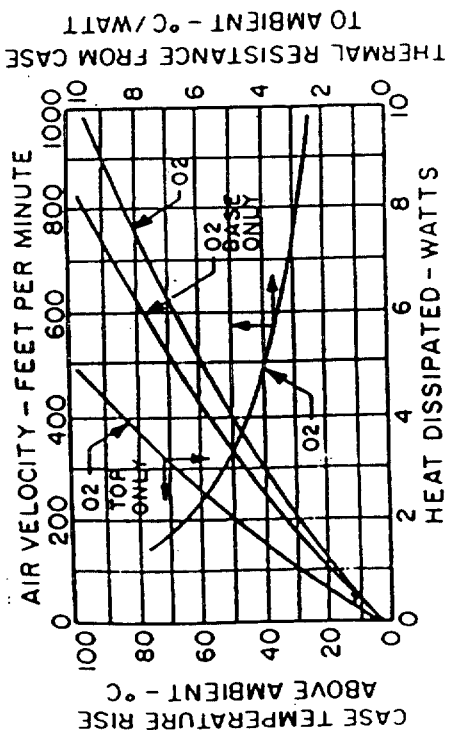
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CONFIGURATION C



CONFIGURATION D



CONFIGURATION B

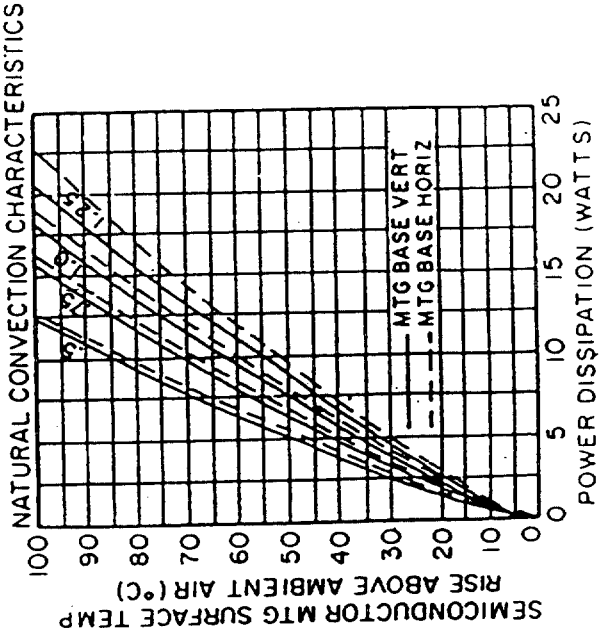
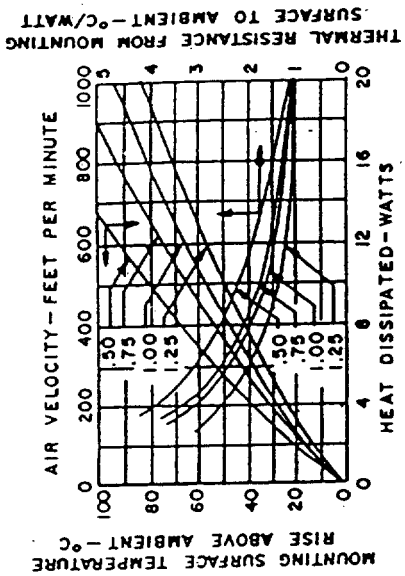
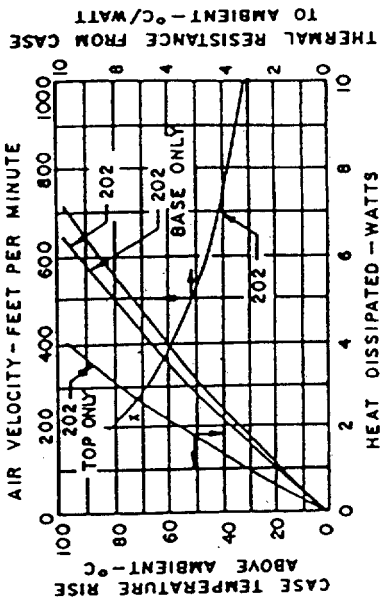


FIGURE 3. Thermal performance characteristics - Continued.

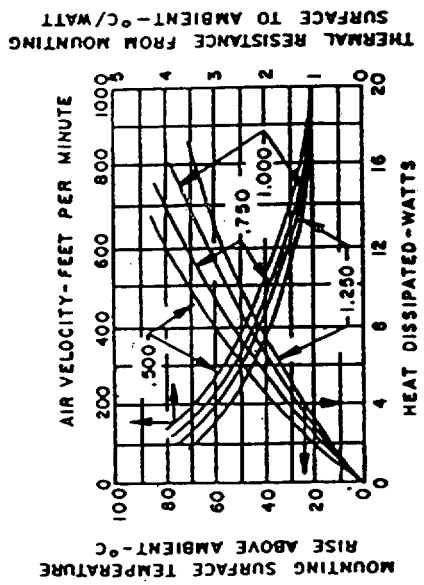
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Check the source to verify that this is the current version before use.



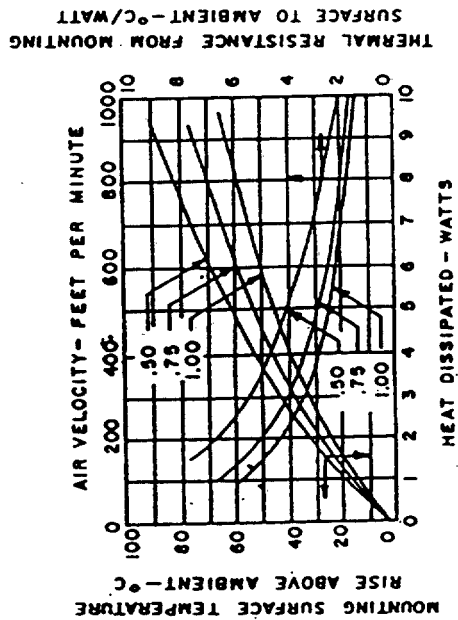
CONFIGURATION F



CONFIGURATION H



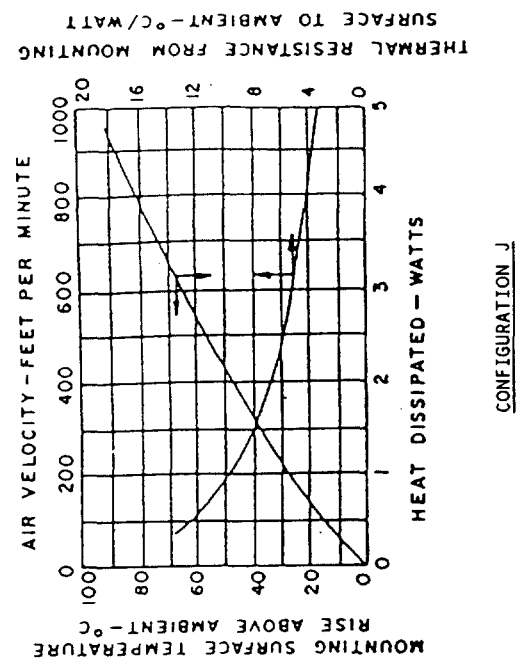
CONFIGURATION E



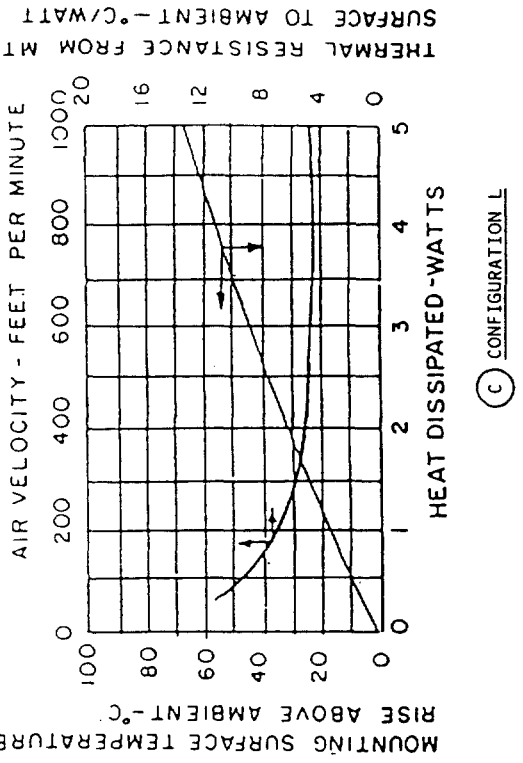
CONFIGURATION G

FIGURE 3. Thermal performance characteristics - Continued.

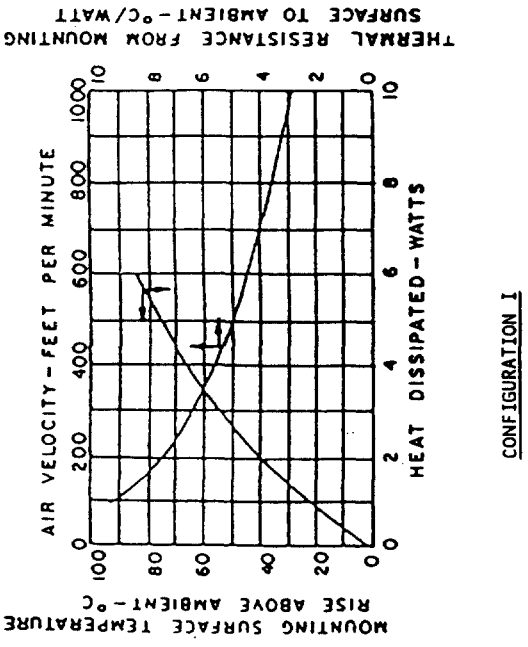
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:49Z
Check the source to verify that this is the current version before use.



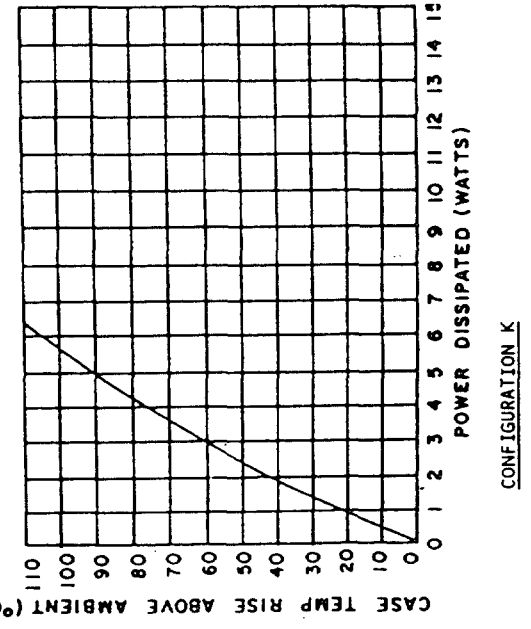
CONFIGURATION J



(C) CONFIGURATION L



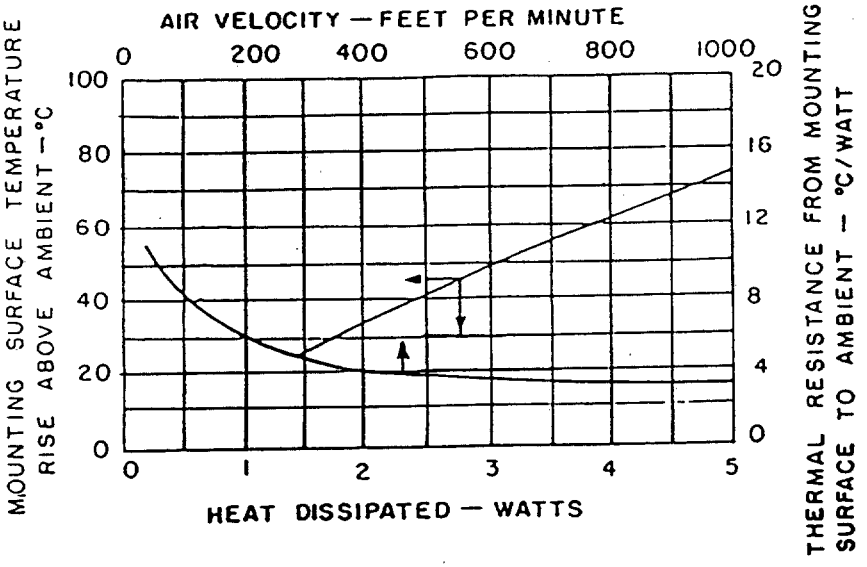
CONFIGURATION I



CONFIGURATION K

FIGURE 3. Thermal performance characteristics - Continued.

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© CONFIGURATION M

FIGURE 3. Thermal performance characteristics - Continued.

REQUIREMENTS:

Configurations and dimensions. See figure 1.

Mounting hole patterns: See figure 2.

JEDEC case size: See figure 2.

Material: Aluminum alloy (class 1)

Finish: Black anodize (A), insulating coating (C), hard anodize (H), or gold chromate (R).

Mounting surface flatness: See figure 1.

Surface roughness: Not applicable.

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Thermal resistance: See table I.

Recommended thermal performance characteristics in forced air and natural convection (as opposed to power dissipation): See figure 3.

(C) Part or Identifying Number (PIN): M87111/05- (plus dash number from table I and finish letter as shown above).

PIN example: M87111/05-1A001.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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